SKILL WEEK 6

Task 1:

Code:

```
#include <stdio.h>

int main() {
char input[200];
    int i;

    printf("Enter a string with escape characters: ");
fgets(input, sizeof(input), stdin);

    printf("\nProcessed string:\n");
for (i = 0; input[i] != '\0'; i++)
{           if (input[i] == '\\' && input[i + 1] !=
'\0') {
             i++;
             switch (input[i])
{                    case
'n':
printf("\n");
break;
case 't':
printf("\t");
break;
case ' ':
printf(" ");
break;
case '\\':
printf("\\");
break;
                case '$':
printf("$");
break;
case '!':
printf("!");
break;
                default:
                  printf("%c", input[i]);
            }
```

```
        } else {            printf("%c",
input[i]);
        }
    }

    return 0;
}
```

```
charith@DESKTOP-RBQQQSS:~$ nano s6.1.c
charith@DESKTOP-RBQQQSS:~$ gcc s6.1.c -o s6.1
charith@DESKTOP-RBQQQSS:~$ ./s6.1
Enter a string with escape characters: Hello World

Processed string:
Hello World
charith@DESKTOP-RBQQQSS:~$
```

Task 2:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {

    pid_t pid;

    pid = fork();
if (pid < 0) {
perror("fork failed");        return
1;
    }

    if (pid == 0) {

        // Child process         printf("Child
process created.\n");         printf("Child
PID: %d\n", getpid());

        char *args[] = {"ls", "-l", NULL};         execvp(args[0], args);

        // This runs only if execvp fails
perror("execvp failed");         exit(1);
```

```
    } else {

        // Parent process          printf("Parent
process.\n");          printf("Parent PID:
%d\n", getpid());

        wait(NULL);

        printf("Child process finished.\n");
    }

    return 0;
}
```

Output:

```
charith@DESKTOP-RBQQQSS:~$ nano s6.2.c
charith@DESKTOP-RBQQQSS:~$ gcc s6.2.c -o s6.2
charith@DESKTOP-RBQQQSS:~$ ./s6.2
Parent process.
Parent PID: 666
Child process created.
Child PID: 667
total 868
drwxr-xr-x 11 charith charith  4096 Aug 18 15:11 OSSP
-rwxr-xr-x  1 charith charith 16568 Aug 31 05:13 Producer_consumer
-rw-r--r--  1 charith charith  3078 Aug 31 05:09 Producer_consumer.c
-rwxr-xr-x  1 charith charith 16440 Aug 31 05:16 Producer_consumer1
-rw-r--r--  1 charith charith  2427 Aug 31 05:15 Producer_consumer1.c
-rwxr-xr-x  1 charith charith 16296 Aug 31 05:06 Task1
-rw-r--r--  1 charith charith  1184 Aug 31 05:05 Task1.c
-rwxr-xr-x  1 charith charith 16384 Aug 31 05:08 Task2
-rw-r--r--  1 charith charith   971 Aug 31 05:07 Task2.c
-rwxr-xr-x  1 charith charith 15952 Jul 28 04:18 a.out
-rwxr-xr-x  1 charith charith 16040 Aug  3 05:53 childp8
-rw-r--r--  1 charith charith   209 Aug  3 05:51 childp8.c
-rwxr-xr-x  1 charith charith 16384 Aug 31 05:28 client
-rw-r--r--  1 charith charith   757 Aug 31 05:24 client.c
prw-r--r--  1 charith charith     0 Aug 31 05:44 client_to_server
-rw-r--r--  1 charith charith     0 Aug 24 06:46 copy.txt
-rwxr-xr-x  1 charith charith 16128 Jul 31 10:00 display
-rw-r--r--  1 charith charith   419 Jul 31 09:59 display.c
-rwxr-xr-x  1 charith charith 16288 Aug 18 16:15 exp2
-rw-r--r--  1 charith charith   969 Aug 18 16:14 exp2.c
-rwxr-xr-x  1 charith charith 16296 Aug 18 16:37 exp3
-rw-r--r--  1 charith charith  1333 Aug 18 16:37 exp3.c
-rwxr-xr-x  1 charith charith 16248 Sep 21 04:19 exp7.1
-rw-r--r--  1 charith charith   840 Sep 21 04:19 exp7.1.c
-rwxr-xr-x  1 charith charith 16296 Sep 21 04:38 exp7.2
```